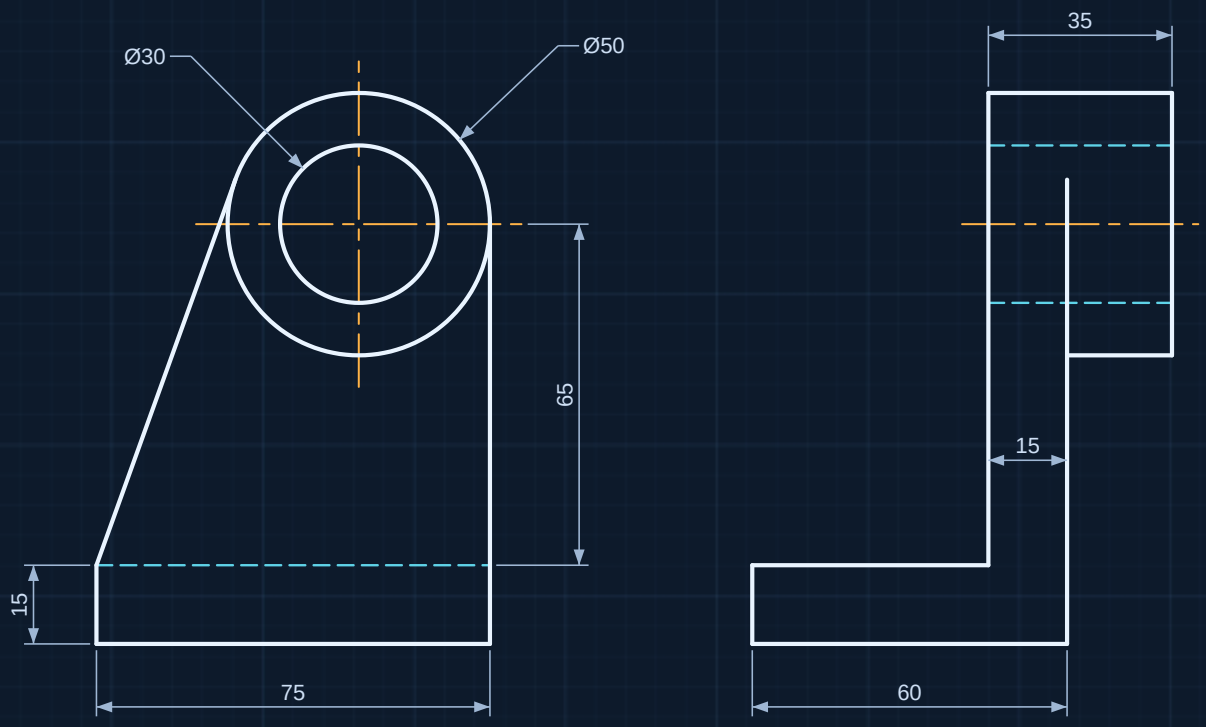
COMPUTER-AIDED DRAWING LAB REPORT

AutoCAD 2020 · EX 3.11

Three Orthographic Views of the Bracket

Exercise **3.11** drawn in **AutoCAD**



PREPARED BY

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01 Introduction and Objectives

What is AutoCAD?

AutoCAD is one of the most widely used Computer Aided Design (CAD) programs, developed by Autodesk. It is used across mechanical, civil, architectural and electrical engineering to produce accurate two- and three-dimensional drawings that can be printed, shared, or sent to a workshop for manufacturing.

In manual drafting the draughtsman relies on a ruler, compass and set squares, and any small mistake means erasing and redrawing. In AutoCAD every line and circle is defined by exact numerical values (coordinates, lengths and angles), so the drawing is perfectly accurate and can be modified, copied or moved in seconds. The program also organises the drawing into **layers**, each with its own colour and linetype, which makes the drawing much easier to read.

This exercise was carried out in AutoCAD 2020. The figure below shows the program interface after the drawing was completed: the Ribbon at the top, the drawing area in the middle and the command line at the bottom.

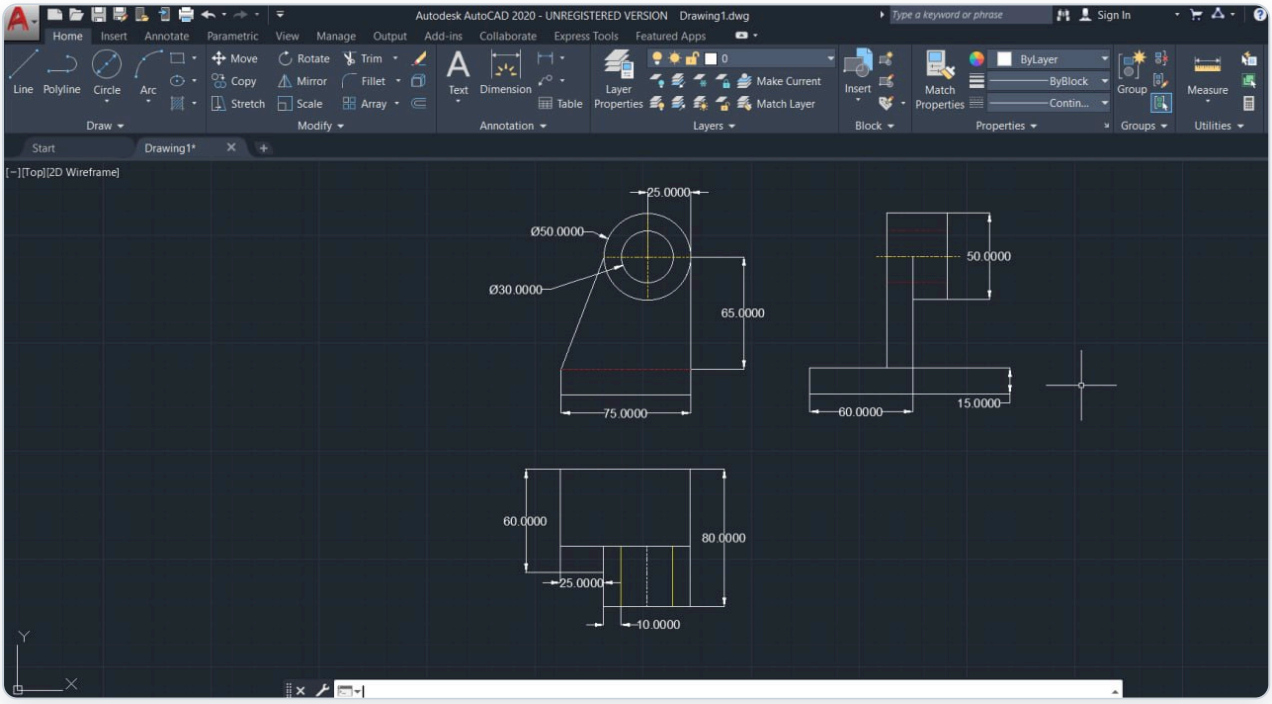


Figure 1. The AutoCAD 2020 interface after completing the three views of the bracket

Parts of the interface

- **Title bar:** shows the program name and the name of the open file (Drawing1.dwg).
- **Ribbon:** contains tabs (Home, Insert, Annotate ...). The Home tab holds the Draw, Modify, Annotation and Layers panels.

- **Drawing area:** the large dark space where the drawing is made, with the X–Y coordinate icon in its lower-left corner.
- **Command line:** where command names and values are typed, and where AutoCAD shows the prompts it needs answered.
- **Status bar:** at the bottom of the screen, containing drafting aids such as Ortho and Object Snap.



Figure 2. The Draw panel: Line, Polyline, Circle and Arc

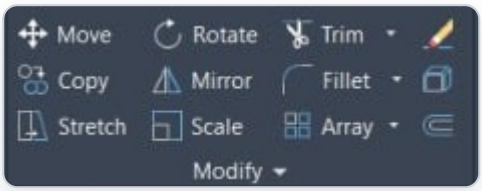


Figure 3. The Modify panel: Move, Copy, Trim, Mirror and more

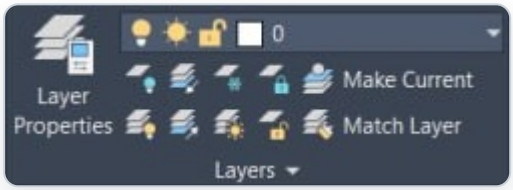


Figure 4. The Layers panel on the Home tab

Objectives of the exercise

- Learn how to set up a new drawing file and define the units of measurement.
- Load linetypes (Hidden, Center) and use colours and layers to organise the drawing.
- Apply the rules of orthographic projection to obtain three views from a pictorial drawing.
- Practise drawing commands (Line, Circle) and modify commands (Trim, Erase, Offset).
- Use object snaps such as Tangent to draw a tangent line accurately.
- Dimension the drawing correctly, clearly and without repetition.

02 Theory of Orthographic Projection

Orthographic projection is a method of representing a three-dimensional object on paper using several two-dimensional views. In each view the object is viewed perpendicular to one of its faces, so only two of its three dimensions are visible. By combining the views, the reader can picture the real shape of the part and find every one of its dimensions.

The three views

- **Front view:** the object seen from the front; it shows the **width and height**. The face with the most detail is usually chosen as the front, and in this exercise it is the face showing the two circles.
- **Side view:** the object seen from the side; it shows the **depth and height**.
- **Top view (plan):** the object seen from above; it shows the **width and depth**.

Relationship between the views

The views are not separate pictures; they are linked by fixed rules:

- The front and side views share the **same height**, so their horizontal lines line up.
- The front and top views share the **same width**, so their vertical lines line up.
- The side and top views share the **same depth**, which is transferred between them using a 45° line.

Because of these relationships, **projection lines** are drawn from the front view to the other two views, so all points stay aligned without recalculating every coordinate. The figure below illustrates the idea: grey lines are projection lines and blue lines are the resulting side and top views.

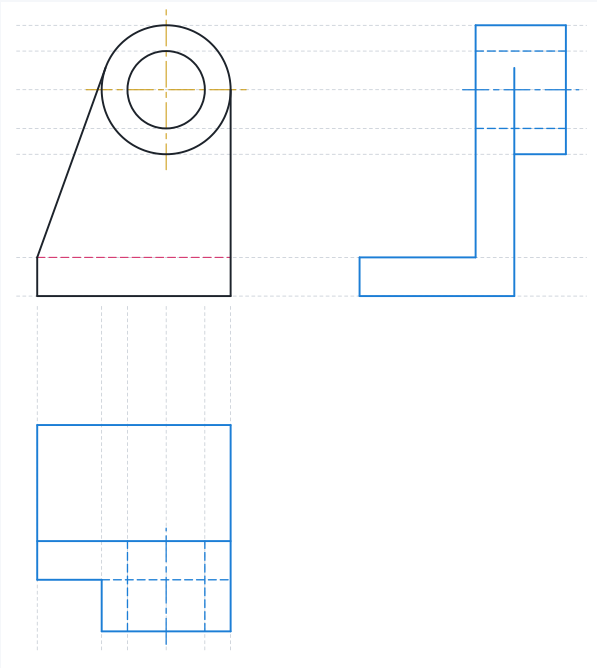


Figure 5. Projection lines connect the front view to the side view (right) and the top view (below)

First-angle projection

This exercise uses **first-angle projection**, the system adopted in our course. In first angle, the **top view is placed below the front view**, and the **left side view is placed to the right of the front view**. That is why the front of the part (the projecting cylinder) appears at the bottom of the top view and on the right-hand side of the side view.

Types of lines

Every line in an engineering drawing has a meaning, and one type must never be used in place of another:

Thick continuous line: visible edges, the edges the viewer can see directly.

Dashed (Hidden) line: edges hidden behind other parts, such as the Ø30 hole in the side and top views.

Centre line: long dash–short dash, used for the axes of circles, cylinders and holes.

Thin continuous line: dimension lines and extension lines.

03 Description of the Part

The figure below is Exercise 3.11 (the bracket) as given in the textbook. It is drawn as an isometric view with the main dimensions marked. Before drawing, the figure must be read carefully to understand the features that make up the part.

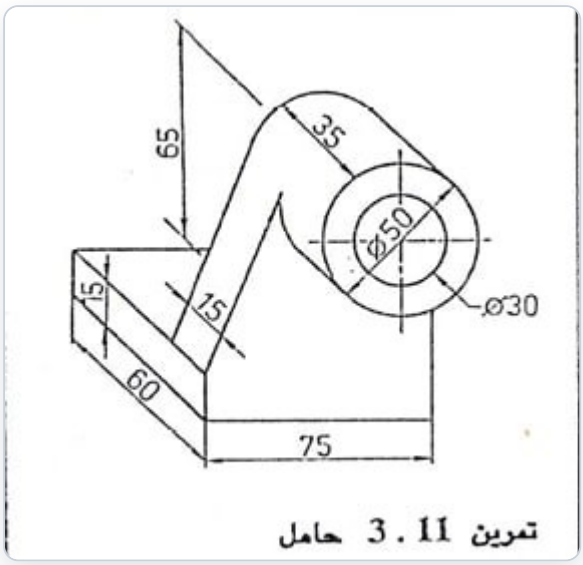


Figure 6. Pictorial view of Exercise 3.11 – the bracket (from the textbook)

Features of the part

- **Base:** a rectangular block 75 long, 60 deep and 15 thick, on which the whole part rests.
- **Web (support rib):** a plate 15 thick rising from the front edge of the base. Its right side is vertical; its left side is inclined, starting at the top-left corner of the base and rising until it is **tangent** to the cylinder.
- **Cylinder (boss):** outside diameter 50, length 35, with a through hole of diameter 30. Its centre is 65 above the top of the base, and it projects forward beyond the base.

Main dimensions shown on the figure:

`75` Base length | `60` Base depth | `15` Base thickness | `15` Web thickness

`65` Centre height above base | `Ø50` Cylinder diameter | `Ø30` Hole diameter | `35` Cylinder length

Calculated values

Some of the values needed for drawing are not written directly on the figure, so they are calculated from the given dimensions before starting. This makes drawing faster and prevents mistakes:

Distance of circle centre from the left edge (cylinder touches the right edge)	`75 - 25 = 50`

Height of circle centre from the ground	`15 + 65 = 80`
Top of the cylinder	`80 + 25 = 105`
Bottom of the cylinder	`80 - 25 = 55`
Height of the hole edges (side view)	`80 ± 15 = 65, 95`
Position of the hole edges from the left (top view)	`50 ± 15 = 35, 65`
Cylinder projection in front of the base	`35 - 15 = 20`
Total depth in the top view	`60 + 20 = 80`

❓ **Why?** The right side of the web is vertical and touches the cylinder on its right. Since it is in line with the right edge of the base, the circle centre lies one radius (25) from it, i.e. 50 from the left edge.

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04 Setting Up the Drawing Environment

Before drawing a single line, the file must be prepared properly: set the units, load the linetypes, create the layers and turn on the drafting aids. This takes only a few minutes but saves a great deal of time and correction later.

1 Start a new drawing

From the Start tab choose **New**, or type the command `NEW`. The **acadiso.dwt** template is preferred because it is based on the metric system (millimetres), whereas **acad.dwt** is based on inches.

2 Set the drawing units (UNITS)

Type `UNITS` or its alias `UN` and press Enter. The **Drawing Units** dialog opens, where the following are set:

- **Length – Type:** Decimal, the system used with millimetres.
- **Length – Precision:** 0.00, two decimal places, which is enough for this drawing.
- **Angle – Type:** Decimal Degrees, with precision 0.00.
- **Clockwise:** left unchecked, so positive angles are measured anticlockwise (the standard direction).
- **Insertion scale:** Millimeters, so that one drawing unit equals one millimetre.

- **Sample Output:** shows an example of coordinates with the chosen settings, such as `1.50,2.00,0.00`.

Then click **OK**.

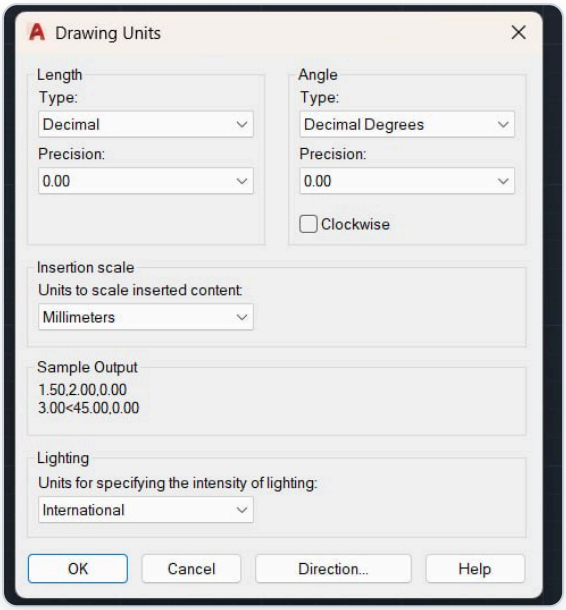


Figure 7. The Drawing Units dialog after setting the options

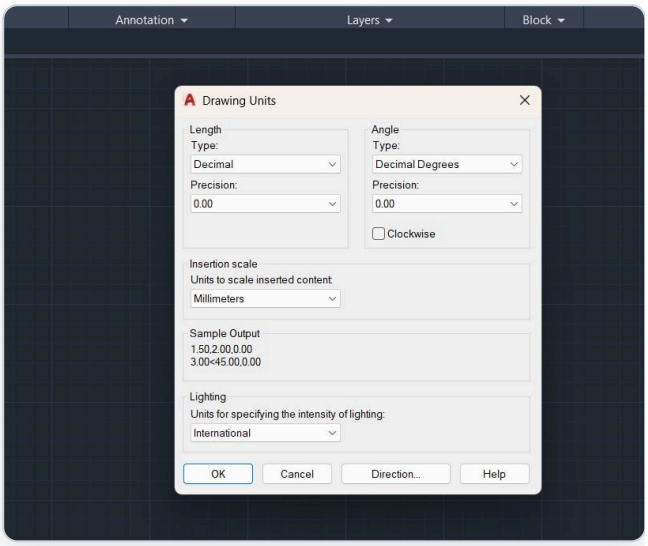


Figure 8. The same dialog shown over the drawing area

```
Command Line
Command: UNITS                                   open the Units dialog
Length Type: Decimal
Precision: 0.00
Insertion scale: Millimeters
 OK
```

3 Load the linetypes (LINETYPE)

A new file only contains the **Continuous** linetype. Type `LINETYPE` or `LT`, or click the Linetype column in the Layer Properties Manager, to open the **Select Linetype** dialog, which shows only one linetype.

Click **Load...** to open the **Load or Reload Linetypes** dialog, which lists all linetypes in the **acad.lin** file. Scroll down and select **HIDDEN** (hidden line), then hold Ctrl and select **CENTER** (centre line), and click OK.

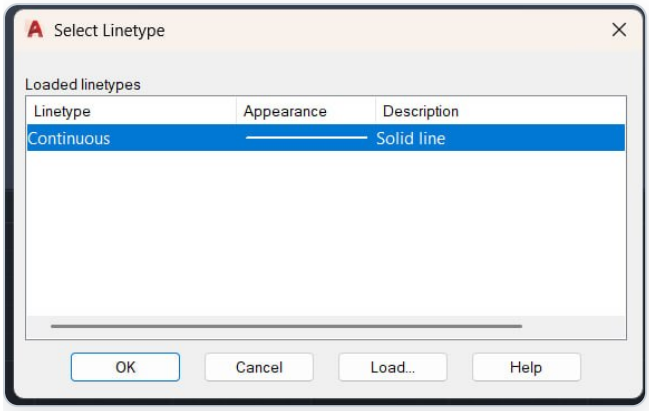


Figure 9. Select Linetype: at first only Continuous is loaded

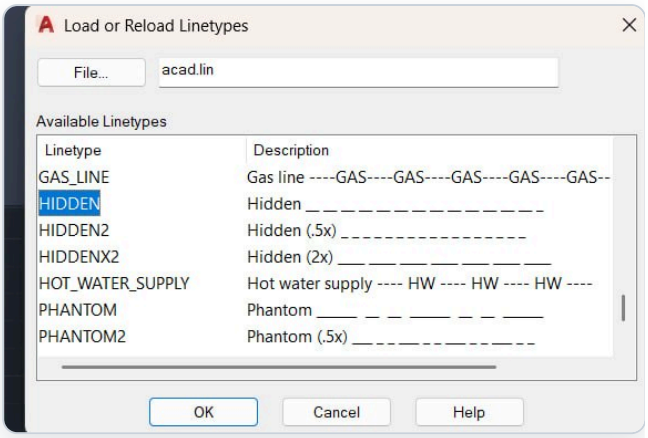


Figure 10. Load or Reload Linetypes with HIDDEN selected

> 💡 **Tip** The list also contains HIDDEN2 and HIDDENX2. The suffix 2 means dashes at half size (0.5x) and X2 means double size (2x). The normal HIDDEN linetype is used here; if the dashes look too small or too large, adjust them with LTSCALE.

4

Choose colours

Clicking the colour box in the Layer Properties Manager opens the **Select Color** dialog. It has three tabs: **Index Color** (AutoCAD's 255 numbered colours), **True Color** (any RGB colour) and **Color Books** (ready-made colour books).

The standard colours in the bottom row have fixed numbers: 1 red, 2 yellow, 3 green, 4 cyan, 5 blue, 6 magenta and 7 white. These are preferred because they are clear on the dark background. Colour 7 (white) prints as black on paper.

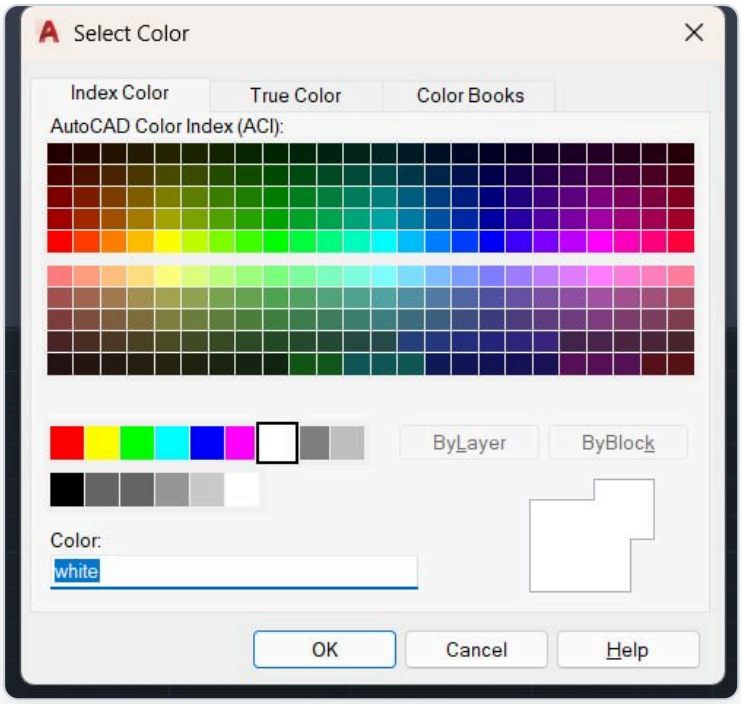


Figure 11. The Select Color dialog and the AutoCAD Color Index (ACI)

5 Create the layers (LAYER)

Type `LA` or click **Layer Properties** on the Layers panel to open the Layer Properties Manager. Click **New Layer**, type the layer name, then set its colour, linetype and lineweight. The following layers were created:

- **Visible:** white (7), Continuous, 0.50 mm — visible edges.
- **Hidden:** red (1), HIDDEN, 0.25 mm — hidden edges.
- **Center:** yellow (2), CENTER, 0.25 mm — centre lines.
- **Dimension:** green (3) or white, Continuous, 0.18 mm — dimensions.

To draw on a layer, select it and click **Set Current**, or pick it from the drop-down list on the Layers panel. The icons beside each layer name are: the light bulb to turn the layer on/off, the sun to freeze/thaw it, and the padlock to lock it against editing.

> 💡 **Tip** With this method there is no need to change the colour or linetype of each line after drawing it; drawing on the correct layer gives the line that layer's properties automatically (ByLayer).

6 Turn on the drafting aids in the status bar

The status bar at the bottom of the screen contains buttons that help to draw accurately. A button is on when it is highlighted in blue:

- **Grid (F7):** displays a background grid to help judge distances.
- **Snap (F9):** makes the cursor jump to grid points.

- **Ortho (F8):** forces lines to be horizontal or vertical only — very important in this exercise.
- **Polar Tracking (F10):** tracks angles at set increments such as 15° or 45°.
- **Object Snap Tracking (F11):** shows temporary alignment paths from snap points.
- **Object Snap (F3):** snaps to exact points on objects such as endpoints, centres and tangent points.

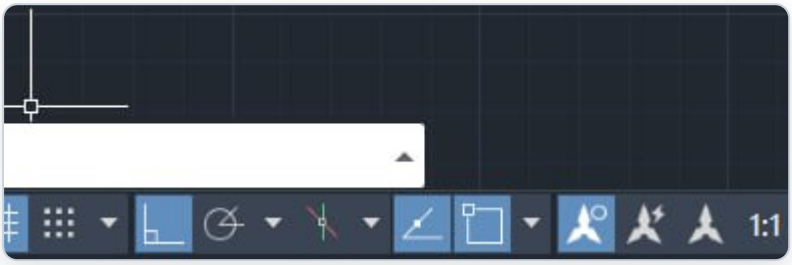


Figure 12. Drafting aids in the status bar (Ortho and Object Snap on, shown in blue)

To set the object snap modes, type `OSNAP` or `OS` and enable: **Endpoint**, **Midpoint**, **Center**, **Quadrant** (the four points on a circle), **Intersection**, **Tangent** and **Perpendicular**.

7 Drawing with Ortho and Dynamic Input

With **Ortho** and **Dynamic Input (F12)** on, a small box next to the cursor shows the length and angle of the line while drawing. Move the cursor in the required direction (up, down, left or right), type the length on the keyboard and press Enter; the line is drawn at exactly that length.

The figure below shows the tooltip `Ortho: 24.45 < 90.00°`, meaning the line is vertical upwards (90°) and its current length is 24.45. At this point the correct length is typed instead.

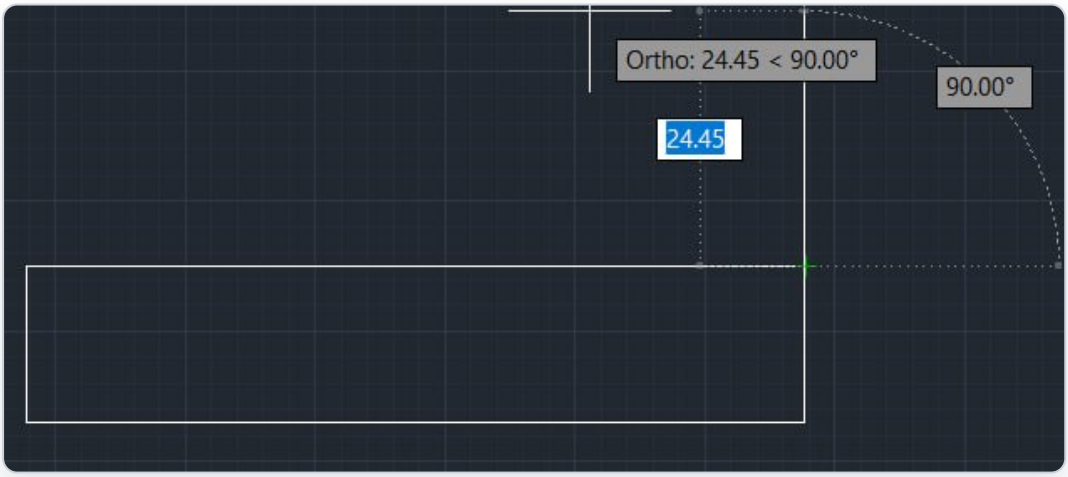


Figure 13. Drawing a vertical line with Ortho, showing length and angle next to the cursor

05 Drawing the Front View

We always start with the front view because it contains the most detail (the two circles and the tangent line), and the heights and widths for the other views are taken from it. The bottom-left corner of the base is taken as the origin `0,0`. In the following figures the **blue** elements are the ones added in each step.

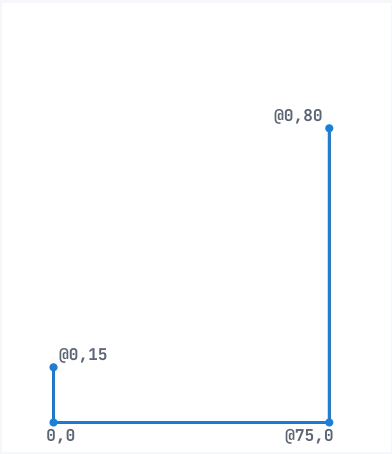


Figure 14. Base and right edge

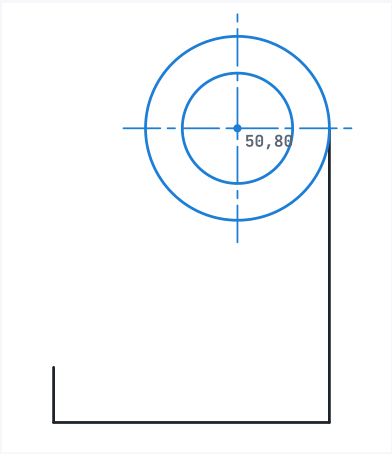


Figure 15. Centre lines and circles

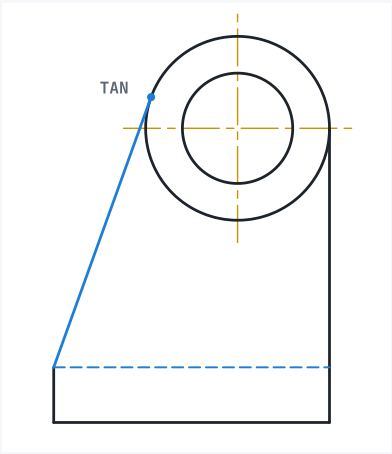


Figure 16. Tangent line and hidden edge

1 Draw the base and the right edge

Make the **Visible** layer current and use the `LINE` (`L`) command. Starting at the origin, draw the bottom edge 75 long, then go up along the right edge 80 (to the level of the circle centre). Relative coordinates start with the `@` symbol and mean "move this distance from the previous point".

```
Command Line
Command: LINE
Specify first point: 0,0                    origin
Specify next point: @75,0                   bottom edge
Specify next point: @0,80                   right edge up to the centre
Specify next point: Enter

Command: LINE
Specify first point: 0,0
Specify next point: @0,15                   left end of the base
Specify next point: Enter
```

2 Draw the centre lines and the two circles

Switch to the **Center** layer and draw the two centre lines, one horizontal and one vertical, crossing at the centre `50,80` and extending about 5 mm beyond the circle on each side. Then return

to the **Visible** layer and draw the circles with `CIRCLE` (`C`): the outer circle with radius 25 (diameter 50) and the inner circle with radius 15 (diameter 30). The **D** option can be used to enter the diameter directly.

```
Command Line
Command: LINE                                          Center layer
Specify first point: 19,80
Specify next point: @62,0                              horizontal axis

Command: LINE
Specify first point: 50,49
Specify next point: @0,62                              vertical axis

Command: CIRCLE                                        Visible layer
Specify center point: 50,80
Specify radius or [Diameter]: 25

Command: CIRCLE
Specify center point: 50,80                            or snap to Center
Specify radius or [Diameter]: D
Specify diameter: 30
```

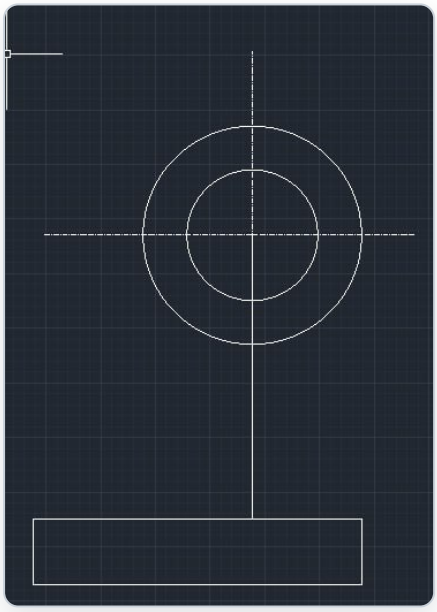


Figure 17. Drawing the two circles and their centre lines above the base

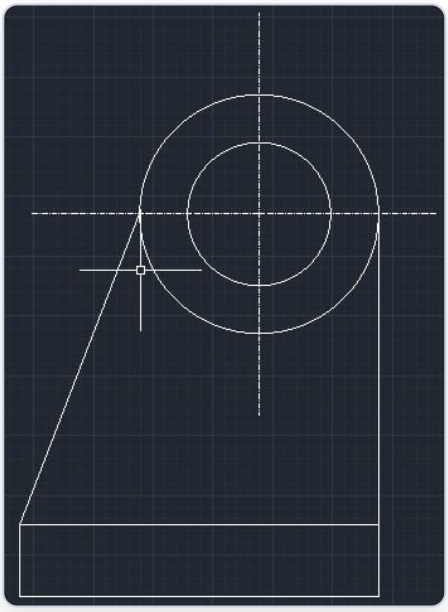


Figure 18. Drawing the tangent line from the corner of the base to the large circle

3 Draw the tangent line

The left side of the web is an inclined line starting at the top-left corner of the base `0,15` and ending where it touches the large circle. The tangent point cannot be found by eye, so the **Tangent** object snap is used: start `LINE`, click the point `0,15`, type `TAN` and move the cursor over the upper-left part of the circle until the tangent marker appears, then click.

```
Command Line
Command: LINE
Specify first point: 0,15                    top-left corner of the base
Specify next point: TAN                      tangent object snap
to (click on circle)                         pick the large circle
Specify next point: Enter
```

❓ **Why?** By calculation the tangent point is at about (26.48, 88.48), slightly above the centre. That is why the line must not be drawn to the left quadrant point of the circle (25, 80) — the line would then not be tangent.

4 The hidden edge in the front view

The web stands on the front edge of the base, so the front face of the web and the front face of the base lie in the same plane and there is no visible line between them. However, the line where the back face of the web meets the top of the base lies behind them, so it is drawn as a hidden line. Switch to the **Hidden** layer and draw a line from `0,15`, 75 long horizontally.

```
Command Line
Command: LINE                                Hidden layer
Specify first point: 0,15
Specify next point: @75,0
Specify next point: Enter
```

06 Drawing the Side View

The side view is placed to the right of the front view and takes **all of its heights** from it. So instead of re-calculating the heights, horizontal projection lines are drawn from the points of the front view towards the right, and the side view is built on them.

1 Draw horizontal projection lines

Use `XLINE` (`XL`) with the `H` (Hor) option to create infinite horizontal lines. Click the key points of the front view using object snaps: the bottom of the base (0), the top of the base (15), the bottom of the circle (55), the lower hole edge (65), the centre (80), the tangent point (88.48), the upper hole edge (95) and the top of the circle (105). These lines can be placed on a temporary layer with a faint colour so they do not mix with the drawing.

```
Command Line
Command: XLINE
Specify a point or [Hor/Ver/Ang/Bisect/Offset]: H
Specify through point: (Endpoint / Quadrant)        click each point
Specify through point: Enter
```

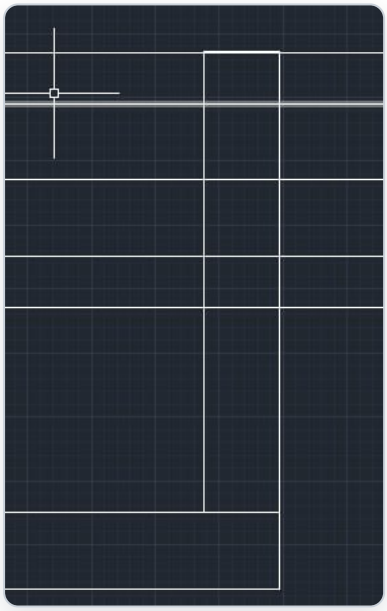


Figure 19. Horizontal projection lines extended from the front view to build the side view

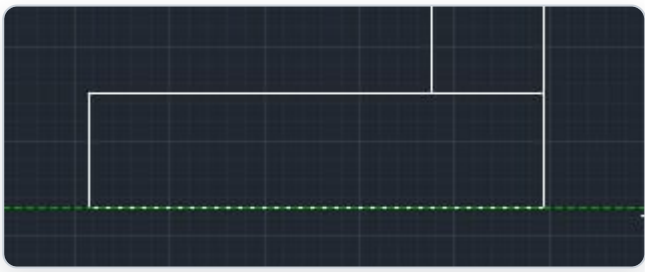


Figure 20. Start of the side view: the base and the beginning of the web

2 Draw the base, web and cylinder

Choose a suitable position for the side view to the right of the front view (for example, the base starts at `X = 125`). This view shows the **depth**: the base is 60 deep and 15 high; the web is 15 thick and stands at the front edge of the base (the right-hand side of this view); the cylinder is

35 long, starting at the back face of the web and projecting 20 forward beyond the base. The top of the cylinder is at 105 and its bottom at 55.

```
Command Line
Command: LINE
Specify first point: 125,0
Specify next point: @60,0                          bottom of the base
Specify next point: @0,88.48                       front face of web up to the tangent line
Specify next point: Enter

Command: LINE
Specify first point: 125,0
Specify next point: @0,15
Specify next point: @45,0                          top of the base
Specify next point: @0,90                          back face of web and cylinder
Specify next point: @35,0                          top of the cylinder
Specify next point: @0,-50                         front face of the cylinder
Specify next point: @-20,0                         underside of the overhang
Specify next point: Enter
```

3 Hidden lines of the hole and the centre line

The Ø30 hole cannot be seen from the side because it is inside the cylinder, so its edges are drawn on the **Hidden** layer: two horizontal lines at heights 65 and 95, the length of the cylinder (35). Then the cylinder axis is drawn on the **Center** layer at height 80, extending slightly beyond both ends.

```
Command Line
Command: LINE                                      Hidden layer
Specify first point: 170,65
Specify next point: @35,0

Command: LINE
Specify first point: 170,95
Specify next point: @35,0

Command: LINE                                      Center layer
Specify first point: 165,80
Specify next point: @45,0
```

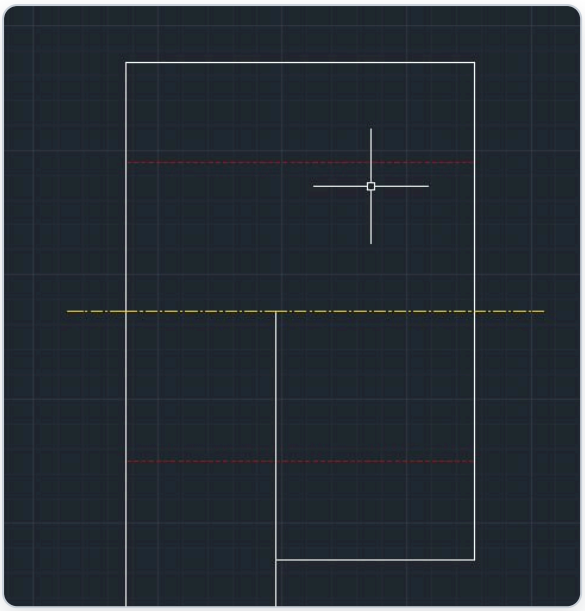


Figure 21. Side view: hidden hole edges (red) and the cylinder axis (yellow)

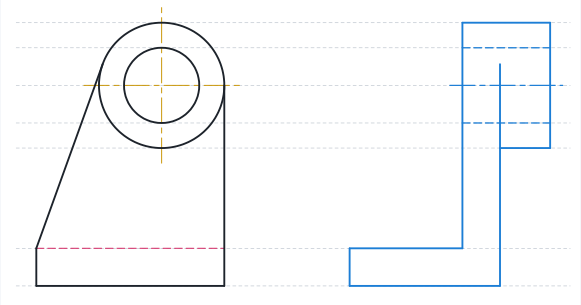


Figure 22. Side view (blue) built on the projection lines

❓ **Why?** In the side view the front face line of the web rises to the height of the tangent point (88.48), not just to the bottom of the cylinder, because the inclined face of the web is closer to the viewer than the cylinder surface in that region.

💡 **Tip** When finished, delete the projection lines with ERASE or turn their layer off, and use TRIM to cut away any extra parts of lines.

07 Drawing the Top View

The top view is placed below the front view. It takes **all widths** (horizontal dimensions) from the front view and the **depth** from the side view. Seen from above, the base appears as a large rectangle and the cylinder as a rectangle projecting from the front (the bottom of the drawing).

1 Draw vertical projection lines

Using `XLINE` with the `V` (Ver) option, draw vertical lines down from the points of the front view: the left edge (0), the start of the cylinder (25), the hole edge (35), the centre (50), the second hole edge (65) and the right edge (75).

The depth is transferred from the side view. The traditional method is to draw a **45° mitre line** in the corner between the side and top views: each vertical line coming down from the side view meets the mitre line and turns into a horizontal line entering the top view, so the depth stays equal in both views. Alternatively, the depth can be entered numerically because it is known (60 for the base and 20 for the overhang).

```
Command Line
Command: XLINE
Specify a point or [Hor/Ver/Ang/Bisect/Offset]: V
Specify through point: (Endpoint / Quadrant)        points of the front view
Specify through point: Enter
```

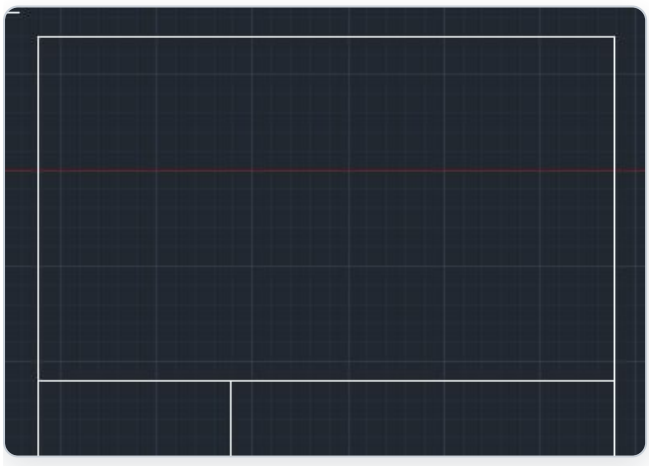


Figure 23. Start of the top view: the base rectangle and the projecting cylinder

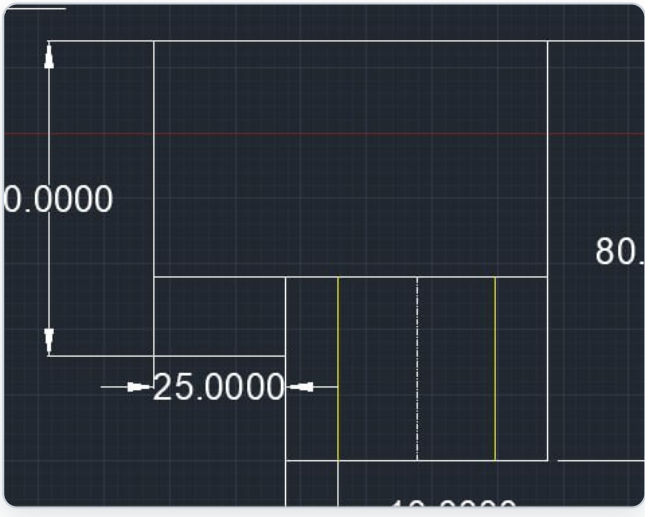


Figure 24. Top view with the hidden hole edges and the centre line

2 Draw the outline

Place the top view below the front view (for example, the front edge of the base at `Y = -110`). Draw the outline as one continuous line: the base 75 × 60, and the cylinder 50 wide, starting 25

from the left and extending 20 below the base. Then draw the back face of the web 15 from the front edge.

```
Command Line
Command: LINE
Specify first point: 0,-110
Specify next point: @0,60                          left edge
Specify next point: @75,0                          back edge
Specify next point: @0,-80                         right edge to end of cylinder
Specify next point: @-50,0                         front of the cylinder
Specify next point: @0,20
Specify next point: @-25,0                         front edge of the base
Specify next point: Enter

Command: LINE
Specify first point: 0,-95
Specify next point: @75,0                          back face of the web
```

3 Hidden lines and the centre line

Seen from above, part of the front edge of the base lies under the projecting cylinder, so it is drawn hidden (from 25 to 75). The edges of the Ø30 hole, 35 and 65 from the left, are also hidden and run the length of the cylinder. Finally, the vertical cylinder axis at 50 is drawn on the **Center** layer.

```
Command Line
Command: LINE                                      Hidden layer
Specify first point: 25,-110
Specify next point: @50,0

Command: LINE
Specify first point: 35,-130
Specify next point: @0,35

Command: LINE
Specify first point: 65,-130
Specify next point: @0,35

Command: LINE                                      Center layer
Specify first point: 50,-135
Specify next point: @0,45
```

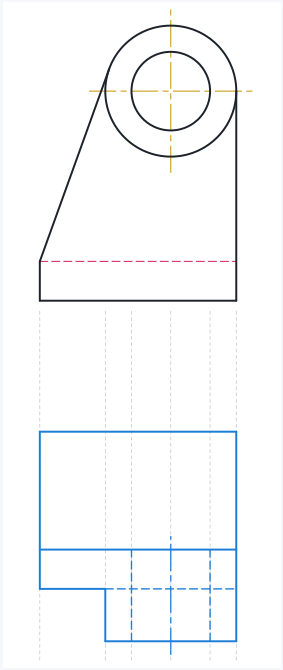


Figure 25. Top view (blue) built on the vertical projection lines from the front view

08 Dimensioning

Dimensions turn the drawing from a picture into a document that can be manufactured, so they must be complete, clear and not repeated. Before dimensioning, the **Dimension** layer is made current and the dimension style is set.

1 Set the dimension style (DIMSTYLE)

Type `DIMSTYLE` or `D` to open the Dimension Style Manager, click **Modify** on the current style and change:

- **Symbols and Arrows tab:** Arrow size = 3.
- **Text tab:** Text height = 4, text placed Above the dimension line.
- **Lines tab:** Extend beyond dim lines ≈ 2 and Offset from origin ≈ 1.
- **Primary Units tab:** Precision = 0, so the value shows as 75 instead of 75.0000.

⚠ **Note** In the screenshots the dimensions show four decimal places, such as 75.0000, because the dimension precision was left at its default. Changing Precision on the Primary Units tab to 0 displays them more simply (75, 60, 15 ...).

2 Dimension commands used

- `DIMLINEAR` / `DLI`: horizontal and vertical dimensions such as 75, 60, 15 and 65. Pick the two points, then drag to place the dimension line.
- `DIMDIAMETER` / `DDI`: the diameter of a circle; click on the circle and the Ø symbol is added automatically (Ø50 and Ø30).
- `DIMRADIUS` / `DRA`: a radius when needed, shown with the letter R.
- `DIMALIGNED` / `DAL`: dimensions parallel to inclined lines.

```
Command Line
Command: DIMLINEAR
Specify first extension line origin: (Endpoint)          start of dimension
Specify second extension line origin: (Endpoint)         end of dimension
Specify dimension line location: (click)                 outside the view
Dimension text = 75

Command: DIMDIAMETER
Select arc or circle: (click circle)
Dimension text = 50
Specify dimension line location: (click)
```

3 Rules for placing dimensions

- Place dimensions **outside** the views wherever possible, between the views and not on top of lines.
- **Do not repeat** a dimension in more than one view; for example the depth 60 appears in one view only.
- Put each dimension in the view where the feature is shown most clearly; diameters go on the circles in the front view.
- Place smaller dimensions closer to the view and larger ones further away, so extension lines do not cross.
- Avoid dimensioning to hidden lines unless necessary.

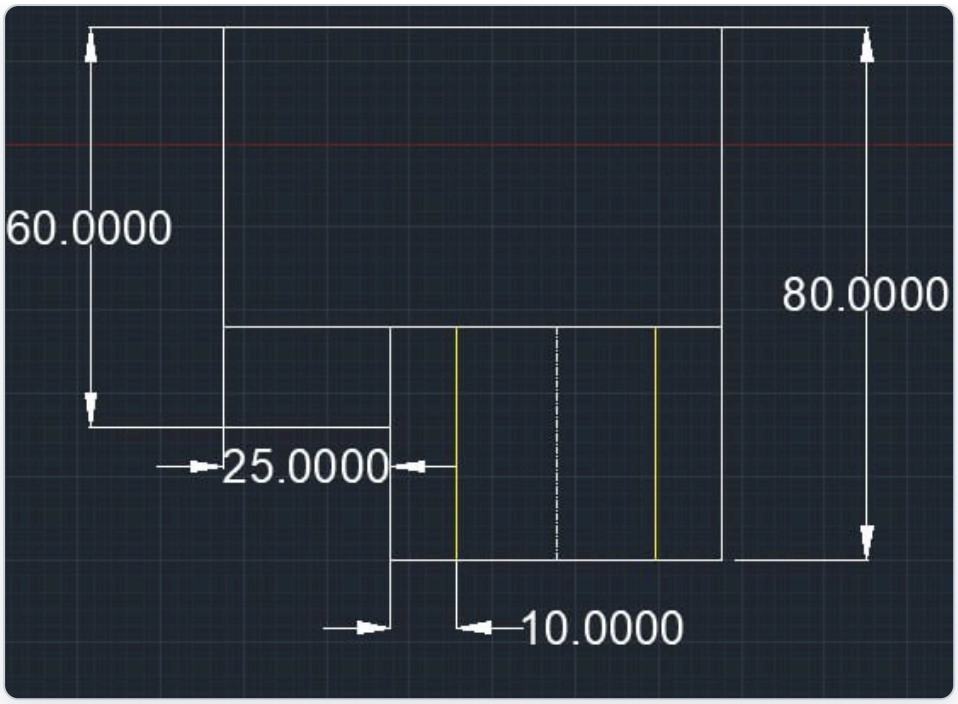


Figure 26. Dimensions on the top view: 60, 80, 25 and 10

09 Final Drawing

After completing the three views and the dimensions, check that the views line up with each other, delete the construction (projection) lines, and make sure every line is on the correct layer. The figure below shows the final drawing as it appeared in AutoCAD:

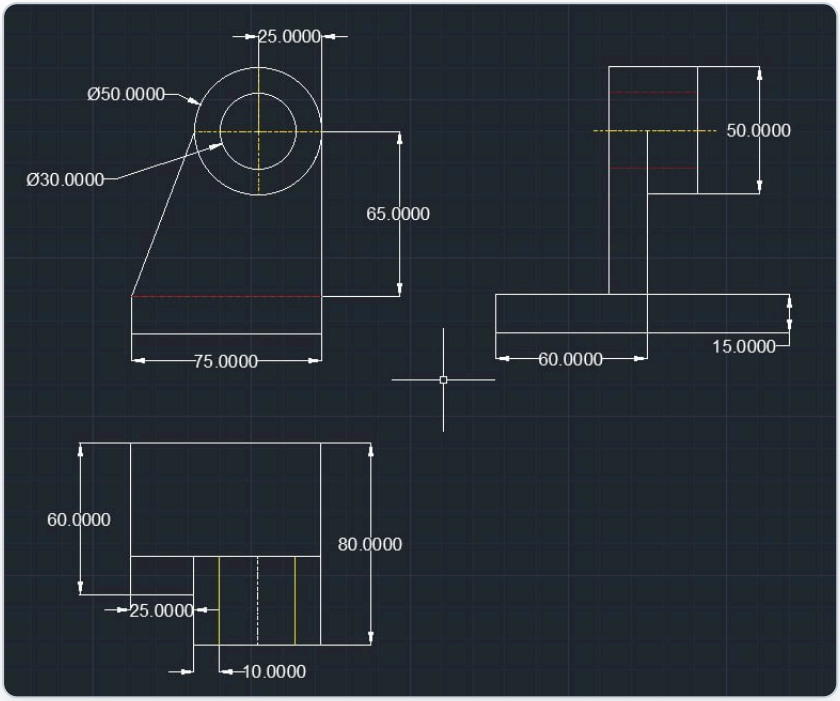
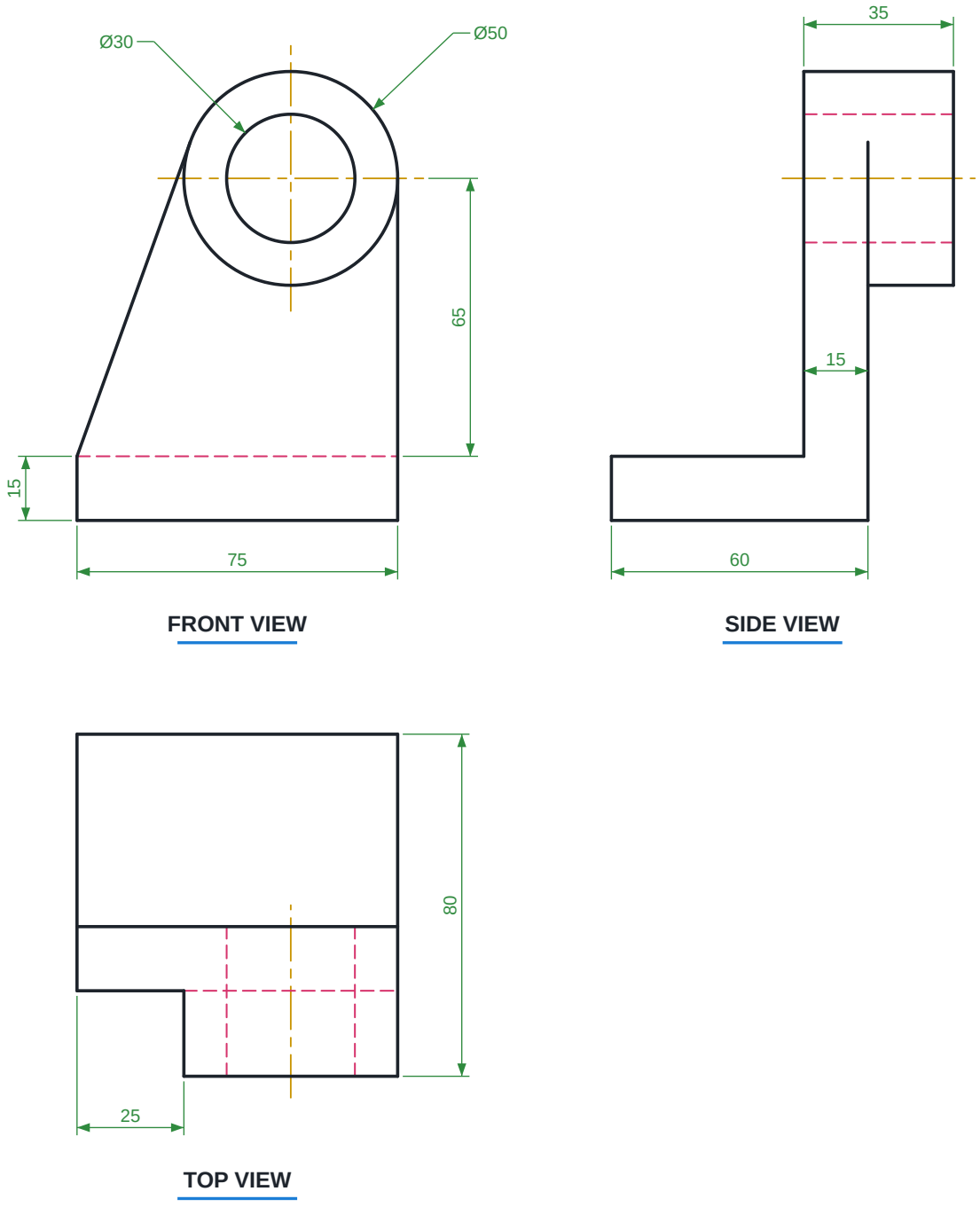


Figure 27. Final drawing in AutoCAD: front view (top left), side view (top right) and top view (bottom)

Note how each view is linked to the others: the heights in the front view (15, 65 and the centre level) line up with the side view, and the widths in the front view (25, 75 and the axis) line up with the top view. The work is saved with `SAVE` or `SAVEAS` in DWG format, and printed with `PLOT` by choosing the printer, A4 paper and a suitable scale.

Final drawing redrawn cleanly for printing



Visible edges Hidden edges Centre lines Dimensions

10 Commands Used and Common Problems

Commands used in the exercise

LINE (L)	Draws straight lines between points; several connected lines can be drawn in one command.
CIRCLE (C)	Draws a circle by centre and radius or diameter; used for the Ø50 and Ø30 circles.
XLINE (XL)	Draws an infinite line (horizontal, vertical or at an angle); used as projection lines between views.
OFFSET (O)	Creates a line parallel to another at a set distance; useful for the 15 web thickness or the hole edges.
TRIM (TR)	Cuts off the parts of lines that extend beyond other boundaries.
ERASE (E)	Deletes selected objects, such as projection lines once they are no longer needed.
MOVE (M)	Moves an object or a whole view to a new position.
MATCHPROP (MA)	Copies the properties of one object (layer, colour, linetype) to another.
ZOOM (Z) / PAN	Zooms and pans the screen without changing the drawing; the mouse wheel can also be used.
LTSCALE (LTS)	Changes the dash scale of all hidden and centre lines in the drawing.
DIMLINEAR (DLI)	Creates horizontal and vertical dimensions.
DIMDIAMETER (DDI)	Creates a diameter dimension with the Ø symbol.

Common problems and their solutions

⚠ **Note Hidden or centre lines appear continuous.** The dash scale does not suit the size of the drawing. Use LTSCALE and try values such as 0.5, 1 or 2 until the dashes are clearly visible.

⚠ **Note The tangent line does not touch the circle exactly.** It was drawn by eye or to a Quadrant point. Delete it and redraw it using the TAN object snap.

⚠ **Note Dimensions show long decimals (75.0000).** Fix it in DIMSTYLE → Modify → Primary Units → Precision = 0.

⚠ **Note The views do not line up.** This happens when each view is drawn separately. Use XLINE projection lines with Ortho and Object Snap Tracking turned on.

⚠ **Note A line was drawn on the wrong layer.** Select it and change its layer from the Layers panel list, or use MATCHPROP to copy the properties of a correct line onto it.

11 Conclusion

Through this exercise I learned that drawing in AutoCAD does not start with lines but with preparation: setting the units, loading the linetypes and creating the layers. This preparation kept the drawing organised, with each line type on its own layer and colour, so any group of lines can be hidden or edited easily.

I also learned the importance of reading the pictorial drawing and calculating the values that are not written on it before starting, such as the position of the circle centre (50, 80) and the overhang of the cylinder (20). Projection lines proved to be the best way of keeping the three views aligned, because they carry heights and widths directly from one view to another without errors.

The object snaps were very useful, especially Tangent, which made the tangent line exact, and Ortho, which kept every horizontal and vertical line perfectly straight. Finally, I learned the rules of dimensioning: dimensions should be outside the views, not repeated, and placed in the view that shows the feature most clearly.

Overall, computer-aided drawing is faster and more accurate than manual drafting, it is easy to modify and can be printed at any scale, which makes it an essential skill for every engineer.

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